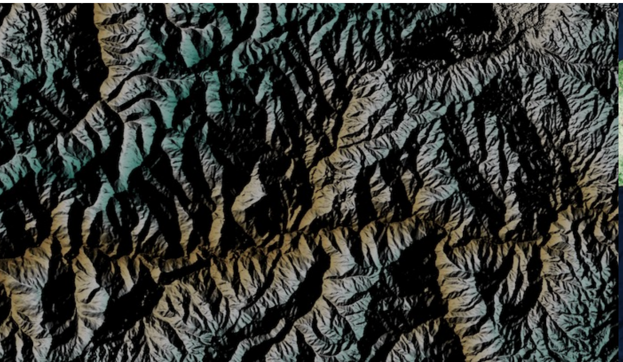
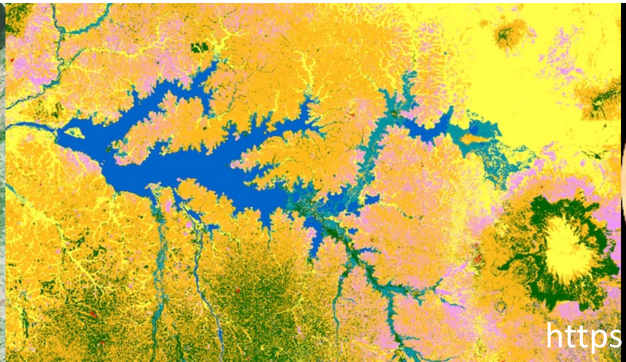
Case studies

- DEM: Copernicus GLO-90 Digital Elevation Model
- NDVI: MODIS Vegetation Indices 16-Day (250m)
- Landcover: ESA WorldCover 2021 or ESA Climate Change Initiative Lanc
- Wind: ERA5 hourly data on single levels






https://planetarycomputer.microsoft.com/catalog

Overview

- Identify the remote sensing product (e.g., Sentinel-2A, MODIS 16-day NDVI)
- Identify a data provider for the product
- Obtain authorization to access the data, if required
- Locate the product files for your region of interest (ROI) and time window
- Download the relevant files or a selected subset
- Merge (mosaic) data from multiple files as needed
- Pre-process data (e.g., compute averages, spectral indices or phenology metrics, smooth spatially, classify)
- Annotation: Extract information at movement (and background) points
- Visualisation and statistical analyses

How to pick environmental predictors

- Resources, conditions, risks
- Movement costs and barriers (wind, topography, fences)

- Know your question
  - Drivers of interest
  - Omitted-variable bias

- Be pragmatic and iterate

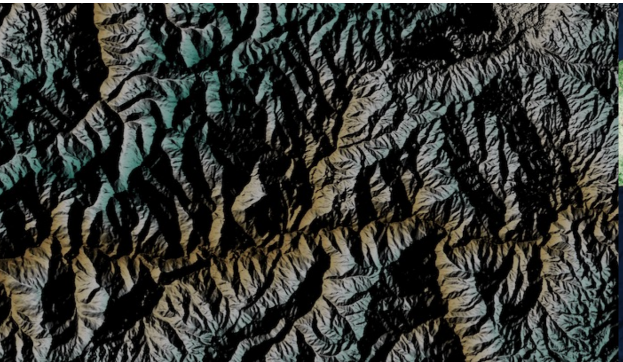
Do-it-yourself remote sensing processing

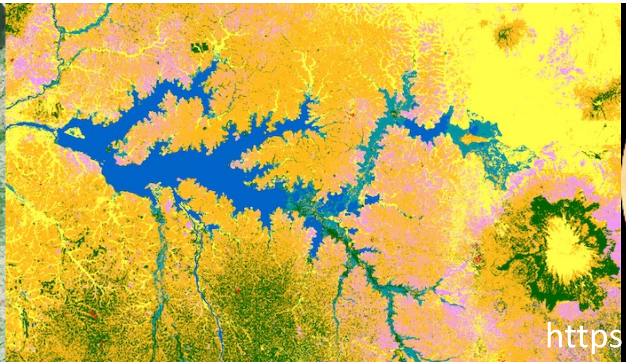
+ Fully tailored to your research question

- Large time investment
- High computational load
- Risk of errors (coding, missing corrections)
- Reduced comparability across studies due to differing environmental representations
- Challenges for reproducibility

Products

- DEM: Copernicus GLO-90 Digital Elevation Model
- NDVI: MODIS Vegetation Indices 16-Day (250m)
- Landcover: ESA WorldCover 2021 or ESA Climate Change Initiative Land
- Wind: ERA5 hourly data on single levels



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Products

- DEM: Copernicus GLO-90 D… Elevation Model
- NDVI: MODIS Veget… indices 16-Day (250m)
- Landcover: ESA Wo… over 2021 or ESA Climate Change Initiative Lanc
- Wind: ERA5 hourly data on single levels

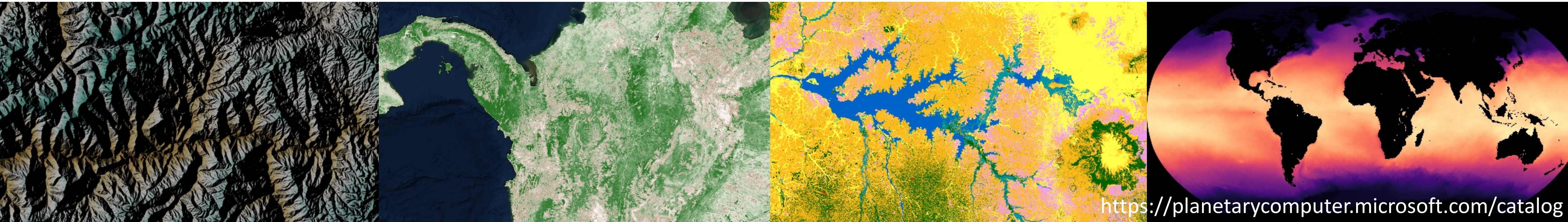
Microsoft Planetary Computer

Copernicus Climate Data Store

https://planetarycomputer.microsoft.com/catalog

https://cds.climate.copernicus.eu/datasets

https://stacspec.org/en/about/datasets



STAC

SpatioTemporal Asset Catalogs

The STAC specification is a **common language to describe geospatial information**, so it can more easily be worked with, indexed, and discovered.

What is STAC

At its core, the SpatioTemporal Asset Catalog (STAC) specification provides a common structure for describing and cataloging spatiotemporal assets.

A *spatiotemporal asset* is any file that represents information about the earth captured in a certain space and time.

> **STAC Item** is the core atomic unit, representing a single spatiotemporal asset as a GeoJSON feature plus datetime and links.
>
> **STAC Catalog** is a simple, flexible JSON file of links that provides a structure to organize and browse STAC Items. A series of best practices helps make recommendations for creating real world STAC Catalogs.
>
> **STAC Collection** is an extension of the STAC Catalog with additional information such as the extents, license, keywords, providers, etc that describe STAC Items that fall within the Collection.
>
> **STAC API** provides a RESTful endpoint that enables search of STAC Items, specified in OpenAPI, following OGC's WFS 3.

https://stacspec.org/en

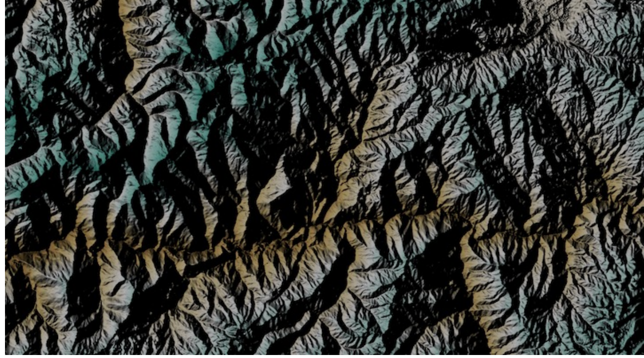
```r
library(rstac)
roi_bbox <- c(31.34, -25.45, 32.29, -23.77) # xmin, ymin, xmax, ymax. epsg:4326
stac_source <- stac("https://planetarycomputer.microsoft.com/api/stac/v1")
stac_query <- stac_search(
  q = stac_source, # Stac query object
  collections = "cop-dem-glo-90", # 90 m global Digital Elevation model
  bbox = roi_bbox # Bounding box
)
executed_stac_query <- get_request(stac_query)
executed_stac_query
```

```
> executed_stac_query
###Items
- features (6 item(s)):
  - Copernicus_DSM_COG_30_S26_00_E032_00_DEM
  - Copernicus_DSM_COG_30_S26_00_E031_00_DEM
  - Copernicus_DSM_COG_30_S25_00_E032_00_DEM
  - Copernicus_DSM_COG_30_S25_00_E031_00_DEM
  - Copernicus_DSM_COG_30_S24_00_E032_00_DEM
  - Copernicus_DSM_COG_30_S24_00_E031_00_DEM
- assets: data, rendered_preview, tilejson
- item's fields:
assets, bbox, collection, geometry, id, links, properties, stac_extensions, stac_version, type
```

```r
library(rstac)
roi_bbox <- c(31.34, -25.45, 32.29, -23.77) # xmin, ymin, xmax, ymax. epsg:4326
stac_source <- stac("https://planetarycomputer.microsoft.com/api/stac/v1")
stac_query <- stac_search(
  q = stac_source, # Stac query object
  collections = "cop-dem-glo-90", # 90 m global Digital Elevation model
  bbox = roi_bbox # Bounding box
)
executed_stac_query <- get_request(stac_query)
# Sign the stac query - this is necessary to actually download the data -
# We do not have the right to download the href directly
signed_stac_query <- items_sign(
  executed_stac_query,
  sign_planetary_computer()
)
# Download the data
assets_download(signed_stac_query, output_dir = my_output_directory,
                asset_names = "data", overwrite = TRUE)
```

```
> # Look at the link to the file with the actual data (first item)
> executed_stac_query$features[[1]]$assets$data$href
[1] "https://elevationeuwest.blob.core.windows.net/copernicus-dem/COP90_hh/Copernicus_DSM_COG_30_S26_00_E032_00_DEM.tif"
> # After signing, the href is changed and a unique identifier is added
> signed_stac_query$features[[1]]$assets$data$href
[1] "https://elevationeuwest.blob.core.windows.net/copernicus-dem/COP90_hh/Copernicus_DSM_COG_30_S26_00_E032_00_DEM.tif?st=2025-12-02T16%3A04%3A23Z&se=2025-12-03T16%3A49%3A23Z&sp=rl&sv=2025-07-05&sr=c&skoid=9c8ff44a-6a2c-4dfb-b298-1c9212f64d9a&sktid=72f988bf-86f1-41af-91ab-2d7cd011db47&skt=2025-12-03T04%3A03%3A02Z&ske=2025-12-10T04%3A03%3A02Z&sks=b&skv=2025-07-05&sig=6DYRvuxiNLkD1ErEb2igINbMk%2Bo%2BRKh7LmNPn%2FXINHM%3D"
```

Download only when necessary



Overview

- Identify the remote sensing product (Copernicus GLO-90 Digital Elevation Model)
- Identify a data provider for the product – Microsoft Planetary Computer
- Obtain authorization to access the data, if required – not required
- Locate the product files for your region of interest (ROI) and time window
- Download the relevant files or a selected subset
- Merge (mosaic) data from multiple files as needed
- Pre-process data (e.g., compute averages, spectral indices or phenology metrics, smooth spatially, classify)
- Annotation: Extract information at movement (and background) points
- Visualisation and statistical analyses

Overview

- Identify the remote sensing product (MODIS 16-day NDVI)
- Identify a data provider for the product (Microsoft Planetary Computer)
- Obtain authorization to access the data, if required (not required)
- Locate the product files for your region of interest (ROI) and time window
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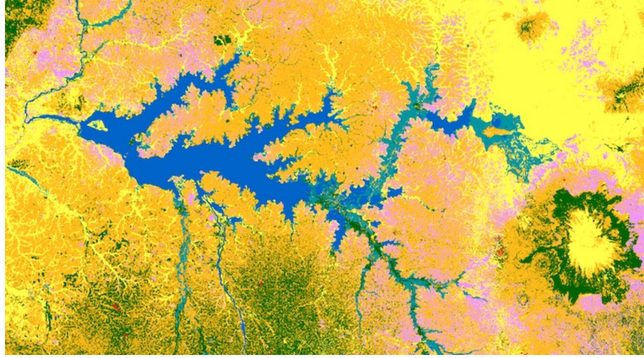

- Identify the remote sensing product (ESA WorldCover 2021)
- Identify a data provider for the product (Microsoft Planetary Computer)
- Obtain authorization to access the data, if required (not required)
- Locate the product files for your region of interest (ROI) and time window
- Download the relevant files or a selected subset
- Merge (mosaic) data from multiple files as needed
- Pre-process data (e.g., compute averages, spectral indices or phenology metrics, smooth spatially, classify, collapse classes)
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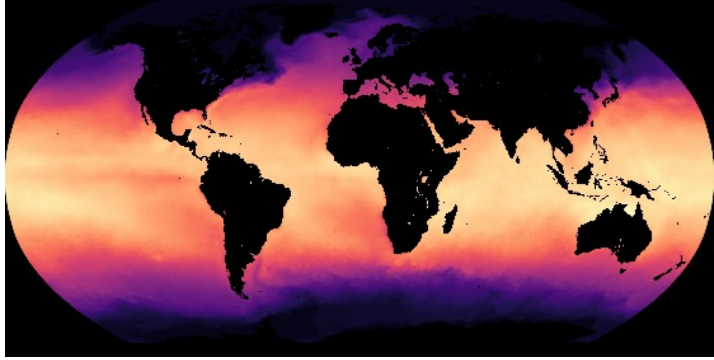
Overview



- Identify the remote sensing product (ERA5 hourly data on single levels)
- Identify a data provider for the product (Copernicus Climate Data Store)
- Obtain authorization to access the data, if required
- Locate the product files for your region of interest (ROI) and time window
- Download the relevant files or a selected subset
- Merge (mosaic) data from multiple files as needed
- Pre-process data (e.g., compute averages, spectral indices or phenology metrics, smooth spatially, classify, collapse classes)
- Annotation: Extract information at movement (and background) points
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